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Power Storage Device and Vehicle

Abstract

A power storage device includes a stacked electrode assembly and a seal member. An electrode plate (bipolar electrode) having a positive electrode layer and a negative electrode layer is stacked between a positive-electrode termination electrode and a negative-electrode termination electrode with a separator being interposed between the electrode plate and each of the positive-electrode termination electrode and the negative-electrode termination electrode so as to form a stacked electrode assembly. An uncoated portion of the negative-electrode termination electrode (current collector) and an uncoated portion of the electrode plate (current collector) extend in a stacking direction of the stacked electrode assembly together with the separator adjacent to each of the uncoated portions, and are folded so as to be stacked on the positive-electrode termination electrode with the separator being interposed between each of the uncoated portions and the positive-electrode termination electrode, thereby forming a crushing-time discharging portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This nonprovisional application is based on Japanese Patent Application No. 2024-027122 filed on Feb. 27, 2024 with the Japan Patent Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND

Field

[0002] The present disclosure relates to a power storage device and a vehicle.

Description of the Background Art

[0003] Japanese Patent Laying-Open No. 2023-152023 discloses a power storage device including a bipolar electrode in which an active material layer is provided on one surface (first surface) and the other surface (second surface) of a current collector. In Japanese Patent Laying-Open No. 2023-152023, a bipolar battery is constituted by a stacked electrode assembly in which bipolar electrodes are stacked with separators being interposed therebetween.

[0004] When a large force is externally applied to the power storage device and the separator of the stacked electrode assembly breaks, a short circuit occurs inside the stacked electrode assembly to generate heat, and the temperature of the power storage device may excessively rise.

SUMMARY

[0005] An object of the present disclosure is to suppress an excessive temperature rise of a power storage device by enabling discharge of the power storage device before a short circuit occurs inside a stacked electrode assembly when large force is applied to the power storage device from outside.

[0006] A power storage device according to an embodiment of the present disclosure includes a positive-electrode termination electrode, a negative-electrode termination electrode, and a stacked electrode assembly. The positive-electrode termination electrode has a current collector having one surface on which a positive electrode active material layer is formed and has an uncoated portion on which the positive electrode active material layer is not formed. The negative-electrode termination electrode has a current collector having one surface on which a negative electrode active material layer is formed and has an uncoated portion on which the negative electrode active material layer is not formed. The stacked electrode assembly has a bipolar electrode stacked between the positive-electrode termination electrode and the negative-electrode termination electrode with a separator being interposed between the bipolar electrode and each of the positive-electrode termination electrode and the negative-electrode termination electrode. The bipolar electrode has a current collector having one surface and the other surface on which the positive electrode active material layer and the negative electrode active material layer are formed respectively, and has an uncoated portion on which each of the positive electrode active material layer and the negative electrode active material layer is not formed. The uncoated portion of one of the positive-electrode termination electrode and the negative-electrode termination electrode extends in a stacking direction of the stacked electrode assembly together with the separator adjacent to the uncoated portion of the one of the positive-electrode termination electrode and the negative-electrode termination electrode, and is folded so as to overlap with the other of the positive-electrode termination electrode and the negative-electrode termination electrode with the

separator being interposed between the uncoated portion of the one of the positive-electrode termination electrode and the negative-electrode termination electrode and the other of the positive-electrode termination electrode and the negative-electrode termination electrode. The uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode extend in the stacking direction of the stacked electrode assembly together with the separator adjacent to each of the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode, and are folded so as to be stacked on the one of the positive-electrode termination electrode and the negative-electrode termination electrode with the separator being interposed between each of the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode and the one of the positive-electrode termination electrode and the negative-electrode termination electrode.

[0007] According to the above configuration, the bipolar battery is formed by the stacked electrode assembly having the bipolar electrode stacked between the positive-electrode termination electrode and the negative-electrode termination electrode with the separator being interposed between the bipolar electrode and each of the positive-electrode termination electrode and the negative-electrode termination electrode. Each of the bipolar electrode, the positive-electrode termination electrode, and the negative-electrode termination electrode has the uncoated portion on which none of the positive electrode active material layer and the negative electrode active material layer is formed. The uncoated portion of the one of the positive-electrode termination electrode and the negative-electrode termination electrode extends in the stacking direction of the stacked electrode assembly together with the separator adjacent to the uncoated portion of the one of the positive-electrode termination electrode and the negative-electrode termination electrode, and is folded so as to overlap with the other of the positive-electrode termination electrode and the negative-electrode termination electrode with the separator being interposed between the uncoated portion of the one of the positive-electrode termination electrode and the negative-electrode termination electrode and the other of the positive-electrode termination electrode and the negative-electrode termination electrode. The uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode extend in the stacking direction of the stacked electrode assembly together with the separator adjacent to each of the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode, and are folded so as to be stacked on the one of the positive-electrode termination electrode and the negative-electrode termination electrode with the separator being interposed between each of the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode and the one of the positive-electrode termination electrode and the negative-electrode termination electrode.

[0008] When the uncoated portion of the one is that of the positive-electrode termination electrode, the uncoated portion of the positive-electrode termination electrode is folded so as to overlap with the negative-electrode termination electrode with the separator being interposed between the uncoated portion of the positive-electrode termination electrode and the negative-electrode termination electrode, and the uncoated portion of the negative-electrode termination electrode and the uncoated portion of the bipolar electrode are folded so as to be stacked on the positive-electrode termination electrode with the separator being interposed between each of the uncoated portion of the negative-electrode termination electrode and the uncoated portion of the bipolar electrode and the positive-electrode termination electrode. In this case, the region in which the uncoated portion of the negative-electrode termination electrode and the uncoated portion of the bipolar electrode are folded so as to be stacked on the positive-electrode termination electrode with the separator being interposed therebetween functions as a crushing-time discharging portion in which the current

collector of the negative-electrode termination electrode and the current collector of the bipolar electrode are stacked with the separator being interposed therebetween.

[0009] When the uncoated portion of the one is that of the negative-electrode termination electrode, the uncoated portion of the negative-electrode termination electrode is folded so as to overlap with the positive-electrode termination electrode with the separator being interposed between the negative-electrode termination electrode and the positive-electrode termination electrode, and the uncoated portion of the positive-electrode termination electrode and the uncoated portion of the bipolar electrode are folded so as to be stacked on the negative-electrode termination electrode with the separator being interposed between each of the uncoated portion of the positive-electrode termination electrode and the uncoated portion of the bipolar electrode and the negative-electrode termination electrode. In this case, the region in which the uncoated portion of the positive-electrode termination electrode and the uncoated portion of the bipolar electrode are folded so as to be stacked on the negative-electrode termination electrode with the separator being interposed therebetween functions as a crushing-time discharging portion in which the current collector of the positive-electrode termination electrode and the current collector of the bipolar electrode are stacked with the separator being interposed therebetween.

[0010] When large force is applied to the power storage device from outside, the separator in the region functioning as the crushing-time discharging portion breaks and a short circuit occurs before the separator inside the stacked electrode assembly breaks and a short circuit occurs, and discharge occurs outside the stacked electrode assembly. Thus, the SOC (State Of Charge) of the stacked electrode assembly is decreased, thereby suppressing an excessive temperature rise of the power storage device.

[0011] A thickness of the separator in a region folded so as to be stacked on the one of the positive-electrode termination electrode and the negative-electrode termination electrode may be thinner than a thickness of the separator in a region included in the stacked electrode assembly.

[0012] According to this configuration, since the thickness of the separator in the region functioning as the crushing-time discharging portion is thinner than the thickness of the separator in the region included in the stacked electrode assembly, the separator in the region functioning as the crushing-time discharging portion is likely to break before the separator inside the stacked electrode assembly breaks when large force is applied to the power storage device from outside. Therefore, an excessive temperature rise of the power storage device can be suppressed more suitably.

[0013] A vehicle according to the present disclosure is a vehicle on which the power storage device is mounted. The power storage device is mounted on a floor of the vehicle. The power storage device may be disposed such that a side of the power storage device on which the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode are folded to be stacked on the one of the positive-electrode termination electrode and the negative-electrode termination electrode with the separator being interposed between the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode faces a road surface.

[0014] According to this configuration, the power storage device is mounted on the floor of the vehicle. Since the side of the power storage device that functions as the crushing-time discharging portion is disposed in the direction facing the road surface, the separator in the region functioning as the crushing-time discharging portion is likely to break before the separator inside the stacked electrode assembly breaks when the floor of the vehicle receives large force due to an interference with the road surface or the like. Therefore, an excessive temperature rise of the power storage device can be suppressed more suitably.

[0015] The foregoing and other objects, features, aspects and advantages of the present disclosure

will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a schematic cross-sectional view of a power storage device according to the present embodiment.

[0017] FIG. 2 is a schematic partial cross-sectional view of the power storage device according to the present embodiment.

[0018] FIG. 3 is a diagram illustrating a function of a crushing-time discharging portion.

[0019] FIG. 4A illustrates a perspective view of a battery module in which a power storage device is housed in an exterior package.

[0020] FIG. 4B illustrates cross-sectional view of a battery module in which a power storage device is housed in an exterior package taken along the line B-B of FIG. 4A.

[0021] FIG. 5A illustrates a perspective view showing a bipolar battery module which is connected in series with the battery module to form a battery assembly.

[0022] FIG. 5B illustrates a cross-sectional view showing a bipolar battery module which is connected in series with the battery module to form a battery assembly taken along the line B-B of FIG. 5A.

[0023] FIG. 6A illustrates a battery pack including a battery module and a bipolar battery module.

[0024] FIG. 6B illustrates a battery pack including a battery module and a bipolar battery module mounted in a vehicle.

DESCRIPTION OF THE EMBODIMENTS

[0025] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings. In the following embodiments, the same or common parts are denoted by the same reference numerals in the drawings, and the description thereof will not be repeated. In the drawings, in order to facilitate understanding of the structure, the ratio may be changed so as to make the structure clear.

[0026] FIG. 1 is a schematic cross-sectional view of a power storage device according to the present embodiment. FIG. 2 is a schematic partial cross-sectional view of the power storage device according to the present embodiment. Note that FIG. 2 is a diagram in which a part of FIG. 1 is removed for ease of explanation using reference numerals.

[0027] Referring to FIG. 2, power storage device 1 includes stacked electrode assembly 10 and seal member 20. The power storage device 1 is, for example, a secondary battery such as a lithium ion battery. The stacked electrode assembly 10 includes a plurality of electrode plates 11, a plurality of separators 15, a positive-electrode termination electrode 16, and a negative-electrode termination electrode 17. The plurality of electrode plates 11, the positive-electrode termination electrode 16, and the negative-electrode termination electrode 17 are stacked with the separator 15 interposed therebetween. The direction in which the stacked electrode assembly 10 is stacked is referred to as a “stacking direction”.

[0028] The separator 15 is formed in a sheet shape. Examples of the separator 15 include a porous film made of a polyolefin-based resin such as polyethylene (PE) or polypropylene (PP), and a woven fabric or a nonwoven fabric made of polypropylene, methyl cellulose, or the like. The separator 15 may be reinforced with a vinylidene fluoride resin compound.

[0029] The plurality of electrode plates 11 are provided between the positive-electrode termination electrode 16 and the negative-electrode termination electrode 17. The electrode plate 11 includes a current collector 12, a positive electrode layer 13, and a negative electrode layer 14. The electrode plate 11 is a bipolar electrode.

[0030] The current collector **12** may contain, for example, at least one selected from the group consisting of aluminum (Al), stainless steel, nickel (Ni), chromium (Cr), platinum (Pt), niobium (Nb), iron (Fe), titanium (Ti), copper (Cu), and zinc (Zn). The current collector **12** may be formed by plating the surface of a metal foil.

[0031] The current collector **12** has a first main surface located on one side in the stacking direction and a second main surface located on the other side in the stacking direction. The negative electrode layer **14** is provided on the first main surface. The positive electrode layer **13** is provided on the second main surface.

[0032] The current collectors **12a** and **12b** of the electrode plate **11** may have a two-layer structure in which a current collector for the positive electrode layer **13** and a current collector for the negative electrode layer **14** are superposed on each other. In this case, the current collector for the positive electrode layer **13** may be made of aluminum (Al), and the current collector for the negative electrode layer **14** may be made of nickel (Ni).

[0033] The positive-electrode termination electrode **16** is located on one side in the stacking direction. The positive-electrode termination electrode **16** includes a current collector **12** and a positive electrode layer **13**. In the positive-electrode termination electrode **16**, the negative electrode layer **14** and the positive electrode layer **13** are not provided on the first main surface located on one side of the current collector **12**, and the positive electrode layer **13** is provided on the second main surface located on the other side of the current collector **12**.

[0034] The negative-electrode termination electrode **17** is located on the other side in the stacking direction. The negative-electrode termination electrode **17** includes a current collector **12** and a negative electrode layer **14**. In the negative-electrode termination electrode **17**, the negative electrode layer **14** is provided on the first type surface located on one side of the current collector **12**, and the negative electrode layer **14** and the positive electrode layer **13** are not provided on the second main surface located on the other side of the current collector **12**.

[0035] The positive electrode layer **13** is formed by applying a positive electrode active material to the second main surface. As the positive electrode active material, for example, a material capable of occluding and releasing charge carriers such as lithium ions can be employed. The positive electrode active material may include olivine type lithium iron phosphate (LiFePO₄). The positive electrode layer **13** corresponds to the “positive electrode active material layer” of the present disclosure.

[0036] The negative electrode layer **14** is formed by applying a negative electrode active material to the first main surface. As the negative electrode parent material, for example, lithium, carbon, a metal compound, an element capable of alloying with lithium, or a compound thereof may be employed. The negative electrode layer **14** corresponds to the “negative electrode active material layer” of the present disclosure.

[0037] In any of the plurality of electrode plates **11**, the negative-electrode termination electrode **17**, and the positive-electrode termination electrode **16**, the peripheral edge portion of the current collector **12** is an uncoated region where the positive electrode layer **13** and the negative electrode layer **14** are not provided. The uncoated region is also referred to as an “uncoated portion”.

[0038] In the present embodiment, two electrode plates **11** (bipolar electrodes) are stacked between the positive-electrode termination electrode **16** and the negative-electrode termination electrode **17** with the separator **15** being interposed therebetween. The stacked electrode assembly **10** includes a first battery cell C1, a second battery cell C2, and a third battery cell C3. The first battery cell C1 includes the negative-electrode termination electrode **17** (the current collector **12d** and the negative electrode layer **14d**), the current collector **12a** and the positive electrode layer **13a** of the electrode plate **11** (the bipolar electrode), and the separator **15a**. The second battery cell C2 includes a current collector **12a** and a negative electrode layer **14a** of the electrode plate **11**, a current collector **12b** and a positive electrode layer **13b** of the electrode plate **11**, and a separator **15b**. The third battery cell C3 includes the positive-electrode termination electrode **16** (the current collector **12c** and the

positive electrode layer **13c**), the current collector **12b** and the negative electrode layer **14b** of the electrode plate **11** (the bipolar electrode), and the separator **15c**. The number of stacked electrode plates **11** may be three or more.

[0039] In the uncoated portion of the current collector **12c** of the positive-electrode termination electrode **16** and the adjacent separator **15c**, an extended portion extending in the stacking direction of the stacked electrode assembly **10** is formed on the left side in FIG. **1**. The extended portion is folded so as to overlap the current collector **12d** of the negative-electrode termination electrode **17**. Thus, the current collector **12c** of the positive-electrode termination electrode **16** overlaps the current collector **12d** of the negative-electrode termination electrode **17** with the adjacent separator **15c** being interposed therebetween.

[0040] In the uncoated portion of the current collector **12b** of the electrode plate **11** adjacent to the positive-electrode termination electrode **16** and the adjacent separator **15c**, an extended portion extending in the stacking direction of the stacked electrode assembly **10** is formed on the right side in FIG. **1**. The extended portion is folded so as to overlap the current collector **12c** of the positive-electrode termination electrode **16**. Thus, the current collector **12b** of the electrode plate **11** overlaps the current collector **12c** of the positive-electrode termination electrode with the adjacent separator **15c** being interposed therebetween.

[0041] In the uncoated portion of the current collector **12a** of the electrode plate **11** adjacent to the negative-electrode termination electrode **17** and the adjacent separator **15b**, an extended portion extending in the stacking direction of the stacked electrode assembly **10** is formed on the right side in FIG. **1**. The extended portion is folded so as to overlap the current collector **12b** of the electrode plate **11**. Thus, the current collector **12a** of the electrode plate **11** overlaps with the current collector **12b** with the adjacent separator **15b** being interposed therebetween.

[0042] In the uncoated portion of the current collector **12d** of the negative-electrode termination electrode **17** and the adjacent separator **15a**, an extended portion extending in the stacking direction of the stacked electrode assembly **10** is formed on the right side in FIG. **1**. The extended portion is folded so as to overlap the current collector **12a** of the electrode plate **11**. Thus, the current collector **12d** of the negative-electrode termination electrode **17** overlaps the current collector **12a** of the electrode plate **11** with the adjacent separator **15a** being interposed therebetween.

[0043] As described above, the uncoated portions of the negative-electrode termination electrode **17** and the electrode plate **11** extend in the stacking direction of the stacked electrode assembly **10** together with the adjacent separators **15**, and are folded so as to be stacked on the positive-electrode termination electrode **16** with the separators **15** being interposed therebetween. A region in which the negative-electrode termination electrode **17** and the uncoated portion of the electrode plate **11** are folded so as to be stacked on the positive-electrode termination electrode **16** with the separator **15** interposed therebetween is also referred to as a “crushing-time discharging portion ES”.

[0044] The seal member **20** is provided so as to seal the periphery of the stacked electrode assembly **10** and the extension portion extending in the stacking direction of the current collector **12** and the separator **15**. The seal member **20** seals an internal space formed between the adjacent electrode plates **11**, an internal space formed between the positive-electrode termination electrode **16** and the electrode plate **11**, and an internal space formed between the negative-electrode termination electrode **17** and the electrode plate **11**. An electrolyte solution is injected into these internal spaces. The seal member **20** is formed by curing a resin member such as a hot melt member, a thermoplastic resin, a thermosetting resin, or a photocurable resin.

[0045] In the power storage device **1** configured as described above, the current collector **12c** of the positive-electrode termination electrode **16** overlapping the current collector **12d** with the separator **15c** being interposed therebetween functions as a positive electrode terminal region. The current collector **12d** of the negative-electrode termination electrode **17** overlapping the current collector **12a** with the separator **15a** being interposed therebetween functions as a negative electrode

terminal region.

[0046] When a large force is externally applied to the power storage device **1** and the separator **15** of the stacked electrode assembly **10** is broken (film ruptured), a short circuit occurs inside the stacked electrode assembly **10** to generate heat, and the temperature of the power storage device **1** may excessively rise.

[0047] In the present embodiment, by forming the crushing-time discharging portion ES, when a large force is applied to the power storage device **1** from the outside, the power storage device **1** can be discharged before a short circuit occurs inside the stacked electrode assembly **10**, and an excessive temperature rise of the power storage device **1** can be suppressed.

[0048] FIG. **3** is a diagram for explaining the function of the crushing-time discharging portion ES. In FIG. **3**, the round bar (cylindrical column) B is pressed against the crushing-time discharging portion ES of the power storage device **1** by the external force F. The crushing-time discharging portion ES is a region in which the negative-electrode termination electrode **17** and the uncoated portion of the electrode plate **11** are folded so as to be stacked on the positive-electrode termination electrode **16** with the separator **15** interposed therebetween. Therefore, as the round bar B progresses due to the external force F, before the separator **15** inside the stacked electrode assembly **10** breaks to cause a short circuit, the separator **15** located in the crushing-time discharging portion ES breaks to cause a short circuit, and discharge is performed outside the stacked electrode assembly **10**.

[0049] For example, with the progress of the round bar B by the external force F, the separator **15a** breaks and a short circuit occurs between the current collector **12d** and the current collector **12a** in the crushing-time discharging portion ES, and the forced discharge of the first battery cell C1 is performed outside the stacked electrode assembly **10**. As a result, the SOC of the first battery cell C1 decreases. When the round bar B advances, the separator **15b** breaks and a short circuit occurs between the current collector **12a** and the current collector **12b** in the crushing-time discharging portion ES, and the forced discharge of the second battery cell C2 is performed outside the stacked electrode assembly **10**. When the round bar B further progresses, the separator **15c** breaks and a short circuit occurs between the current collector **12b** and the current collector **12c** in the crushing-time discharging portion ES, and the forced discharge of the third battery cell C3 is performed outside the stacked electrode assembly **10**.

[0050] As described above, when a large force is applied to the power storage device **1** from the outside, before the separator **15** inside the stacked electrode assembly **10** breaks to cause a short circuit, the separator **15** in the region of the crushing-time discharging portion ES breaks to cause a short circuit, and discharge is performed outside the stacked electrode assembly **10**. As a result, the SOC of the stacked electrode assembly **10** decreases, and thus an excessive temperature rise of the power storage device **1** can be suppressed.

[0051] In the present embodiment, as shown in FIG. **1**, the separator **15** is configured such that the thickness **t2** of the region of the crushing-time discharging portion ES is thinner than the thickness **t1** of the region included in the stacked electrode assembly **10**. Since the thickness **t2** is smaller than the thickness **t1**, before the separator **15** inside the stacked electrode assembly **10** breaks and a short circuit occurs when a large force is applied to the power storage device **1** from the outside, the separator **15** in the region of the crushing-time discharging portion ES easily breaks, and discharge outside the stacked electrode assembly **10** can be performed more reliably.

[0052] Each of FIGS. **4A** and **4B** is a diagram illustrating the battery module **100** in which the power storage device **1** is housed in the exterior package **40**. The battery module **100** is formed by housing the power storage device **1** in the exterior package **40**. FIG. **4A** is a perspective view of the battery module **100**, and FIG. **4B** is a cross-sectional view taken along line B-B in FIG. **4A**. The exterior package **40** includes an exterior package A **41** and an exterior package B **42**. The A exterior package **41** and the B exterior package **42** seal the power storage device **1** inside by joining their peripheral edge portions to each other. The A exterior package **41** is disposed on the current

collector **12d** of the negative-electrode termination electrode **17** located on one side in the stacking direction (on the crushing-time discharging portion ES side). The B exterior package **42** is disposed on the current collector **12c** of the positive-electrode termination electrode **16** located on the other side in the stacking direction.

[0053] The A exterior package **41** includes the first conductive plate **18**, the resin layer **50**, and the first sheet member **31**. The B exterior package **42** includes the second conductive plate **19**, the resin layer **50**, and the second sheet member **32**.

[0054] The first conductive plate **18** and the second conductive plate **19** are provided so as to sandwich the power storage device **1** in the stacking direction. The first conductive plate **18** is disposed on the current collector **12d** of the negative-electrode termination electrode **17**. The first conductive plate **18** is electrically connected to the negative-electrode termination electrode **17** by being disposed in contact with the current collector **12d**. The first conductive plate **18** functions as a negative terminal of the battery module **100** by being electrically connected to the negative-electrode termination electrode **17**. The second conductive plate **19** is disposed on the current collector **12c** of the positive-electrode termination electrode **16**. The second conductive plate **19** is electrically connected to the positive-electrode termination electrode **16** by being disposed in contact with the current collector **12c**. The second conductive plate **19** functions as a positive electrode terminal of the battery module **100** by being electrically connected to the positive-electrode termination electrode **16**. In the battery module **100**, the current can be extracted to the outside from the power storage device **1** accommodated therein via the first conductive plate **18** functioning as the negative electrode terminal and the second conductive plate **19** functioning as the positive electrode terminal without using a tab for extracting the current to the outside.

[0055] The first sheet member **31** constitutes a peripheral edge portion of the exterior package A **41**. The first sheet member **31** is bonded to the peripheral edge of the first conductive plate **18**. In the present embodiment, the first sheet member **31** is bonded to the first conductive plate **18** in a state where the resin layer **50** is interposed between the first sheet member and the peripheral edge of the first conductive plate **18**.

[0056] The second sheet member **32** constitutes a peripheral edge portion of the B exterior package **42**. The second sheet member **32** is bonded to the peripheral edge of the second conductive plate **19**. In the present embodiment, the second sheet member **32** is bonded to the second conductive plate **19** in a state where the resin layer **50** is interposed between the second sheet member and the peripheral edge of the second conductive plate **19**.

[0057] The peripheral edges of the first conductive plate **18** and the second conductive plate **19** are located on the seal member **20**. A resin material having an insulating property may be used for the resin layer **50**. As the resin layer **50**, a resin material that can be welded to the first conductive plate **18** or the second conductive plate **19** may be used. As the resin layer **50**, for example, a heat-fusible resin such as polyethylene, polypropylene, modified polyethylene, or modified polypropylene can be employed.

[0058] In the present embodiment, the first sheet member **31** and the second sheet member **32** are laminate films. The first sheet member **31** includes a first metal layer **310**, a first insulating layer **311**, and a second insulating layer **312**. As the first metal layer **310**, for example, a metal foil such as an Al foil, a Ni foil, a Cu foil, or a stainless steel foil can be used. The first insulating layer **311** and the second insulating layer **312** may be made of a heat-sealable resin such as polyethylene, polypropylene, modified polyethylene, or modified polypropylene. The second insulating layer **312** may be a member different from the first insulating layer **311**. As the second insulating layer **312**, for example, a single layer of polyethylene terephthalate or nylon, or a stacked layer thereof may be employed. The first insulating layer **311** and the second insulating layer **312** may be formed of one layer or may have a multilayer structure.

[0059] The second sheet member **32** includes a second metal layer **320**, a first insulating layer **321**, and a second insulating layer **322**. The second metal layer **320**, the first insulating layer **321**, and

the second insulating layer **322** may be the same as those of the first metal layer **310**, the first insulating layer **311**, and the second insulating layer **312**.

[0060] The first sheet member **31** of the A exterior package **41** and the outer peripheral portion of the second sheet member **32** of the B exterior package **42** are joined by welding. Thus, the power storage device **1** is sealed in the A exterior package **41** and the B exterior package **42** by welding the first insulating layer **311** of the A exterior package **41** and the first insulating layer **321** of the B exterior package **42** in a state of facing each other. A welded portion **70** is formed in an outer peripheral portion where the A exterior package **41** and the B exterior package **42** are welded.

[0061] Each of FIGS. **5A** and **5B** is a diagram showing a bipolar battery module **100A** that is connected in series with the battery module **100** of the present embodiment to form a battery assembly. FIG. **5A** is a perspective view of the bipolar battery module **100A**, and FIG. **5B** is a cross-sectional view taken along line B-B shown in FIG. **5A**. The bipolar battery module **100A** has substantially the same configuration as that of the battery module **100** of FIGS. **4A** and **4B** except that the crushing-time discharging portion **ES** is omitted. The bipolar battery module **100A** includes a power storage unit **1A** having a stacked electrode assembly **10A** and a seal member **20A**, and an exterior package **40A** that accommodates the power storage unit **1A**.

[0062] The stacked electrode assembly **10A** includes a plurality of electrode plates **11A**, a plurality of separators **15A**, a positive-electrode termination electrode **16A**, and a negative-electrode termination electrode **17A**. The plurality of electrode plates **11A**, the positive-electrode termination electrode **16A**, and the negative-electrode termination electrode **17A** are stacked in the stacking direction with the separator **15A** interposed therebetween. The number of the electrode plates **11A** is any number, and may be, for example, 5, 20, or 30.

[0063] The electrode plate **11A** is substantially the same as the electrode plate **11**, and a positive electrode layer **13A** is formed on one surface of the current collector **12A**, and a negative electrode layer **14A** is formed on the other surface of the current collector **12A**. The current collector **12A**, the positive electrode layer **13A**, and the negative electrode layer **14A** are substantially the same as the positive electrode layer **13** and the negative electrode layer **14** described above, and the electrode plate **11A** is a bipolar electrode.

[0064] The separator **15A**, the positive-electrode termination electrode **16A**, and the negative-electrode termination electrode **17A** are substantially the same as the separator **15**, the positive-electrode termination electrode **16**, and the negative-electrode termination electrode **17**, respectively.

[0065] In any of the plurality of electrode plates **11A**, the positive-electrode termination electrode **16A**, and the negative-electrode termination electrode **17A**, the uncoated portion of the peripheral edge portion of the current collector **12A** does not have an extended portion extending in the stacking direction of the stacked electrode assembly **10**. Also, the separator **15A** is not provided with an extension portion extending in the stacking direction of the stacked electrode assembly **10**.

[0066] The seal member **20A** is provided so as to seal the periphery of the stacked electrode assembly **10A**. The seal member **20A** seals an internal space formed between the adjacent electrode plates **11A**, an internal space formed between the positive-electrode termination electrode **16A** and the electrode plate **11A**, and an internal space formed between the negative-electrode termination electrode **17A** and the electrode plate **11A**. An electrolyte solution is injected into these internal spaces. The exterior package **40A** includes an exterior package **41A** and an exterior package **42A**. The exterior package **41A** includes a conductive plate **19A**, a resin layer **50A**, and a sheet member **31A**. The exterior package **42A** includes a conductive plate **18A**, a resin layer **50A**, and a sheet member **32A**.

[0067] The conductive plate **18A** and the conductive plate **19A** are provided so as to sandwich the stacked electrode assembly **10A** in the stacking direction. The conductive plate **19A** is electrically connected to the positive-electrode termination electrode **16A** by being disposed on and in contact with the current collector **12A** of the positive-electrode termination electrode **16A**. The conductive

plate **19A** functions as a positive electrode terminal of the bipolar battery module **100A**. The conductive plate **18A** is electrically connected to the negative-electrode termination electrode **17A** by being disposed in contact with the current collector **12A** of the negative-electrode termination electrode **17A**. The conductive plate **18A** functions as a negative electrode terminal of the bipolar battery module **100A**. Since the other configuration of the exterior package **40A** is the same as that of the exterior package **40** described above, the description thereof will be omitted.

[0068] Each of FIGS. **6A** and **6B** is a diagram illustrating a battery pack BT including the battery module **100** and the bipolar battery module **100A**. As illustrated in FIG. **6A**, the battery module **100** and the bipolar battery module **100A** are stacked between the second conductive plate **19** (positive electrode terminal) of the battery module **100** and the conductive plate **18A** (negative electrode terminal) of the bipolar battery module **100A** with a conductive member **200** being interposed therebetween. The bipolar battery module **100A** is stacked between the conductive plate **18A** (positive electrode terminal) and the conductive plate **19A** (negative electrode terminal) with the conductive member **200** being interposed therebetween. The number of bipolar battery modules **100A** may be any number, three, or ten. As illustrated in FIG. **6A**, a battery pack BT is formed by connecting the battery module **100** and the bipolar battery module **100A** in series to form a battery assembly and housing the battery assembly in a battery case (not illustrated).

[0069] As illustrated in FIG. **6B**, the battery pack BT is mounted on the vehicle V and is disposed on the floor **500** of the vehicle V. The battery pack BT is disposed on the floor **500** of the vehicle V so that the first conductive plate **18** of the battery module **100** faces the road surface G (so that the first conductive plate **18** is vertically downward). Thus, the battery pack BT is mounted on the vehicle V such that the crushing-time discharging portion ES of the power storage device **1** faces the road surface G. Therefore, when the floor **500** of the vehicle V receives a large force due to road surface interference or the like, in the battery module **100**, before the separator **15** inside the stacked electrode assembly **10** breaks and a short circuit occurs, the separator **15** in the region of the crushing-time discharging portion ES breaks and a short circuit occurs, and discharge is performed outside the stacked electrode assembly **10**. As a result, the SOC of the stacked electrode assembly **10** decreases, and thus an excessive temperature rise of the battery pack BT (power storage device **1**) can be suppressed.

[0070] In the above-described embodiment, in the power storage device **1**, the negative-electrode termination electrode **17** and the uncoated portion of the electrode plate **11** extend in the stacking direction of the stacked electrode assembly **10** together with the adjacent separators **15**, and are folded so as to be stacked on the positive-electrode termination electrode **16** with the separators **15** being interposed therebetween. A region in which the negative-electrode termination electrode **17** and the uncoated portion of the electrode plate **11** are folded so as to be stacked on the positive-electrode termination electrode **16** with the separator **15** interposed therebetween is used as the crushing-time discharging portion ES.

[0071] However, the uncoated portions of the positive-electrode termination electrode **16** and the electrode plate **11** may extend in the stacking direction of the stacked electrode assembly **10** together with the adjacent separators **15**, and may be folded so as to be stacked on the negative-electrode termination electrode **17** with the separators **15** being interposed therebetween. In this case, a region in which the uncoated portions of the positive-electrode termination electrode **16** and the electrode plate **11** are folded so as to be stacked on the negative-electrode termination electrode **17** with the separator **15** interposed therebetween functions as the crushing-time discharging portion ES.

[0072] Although the present disclosure has been described and illustrated in detail, it is clearly understood that the same is by way of illustration and example only and is not to be taken by way of limitation, the scope of the present disclosure being interpreted by the terms of the appended claims.

Claims

1. A power storage device comprising: a positive-electrode termination electrode that has a current collector having one surface on which a positive electrode active material layer is formed and that has an uncoated portion on which the positive electrode active material layer is not formed; a negative-electrode termination electrode that has a current collector having one surface on which a negative electrode active material layer is formed and that has an uncoated portion on which the negative electrode active material layer is not formed; and a stacked electrode assembly that has a bipolar electrode stacked between the positive-electrode termination electrode and the negative-electrode termination electrode with a separator being interposed between the bipolar electrode and each of the positive-electrode termination electrode and the negative-electrode termination electrode, wherein the bipolar electrode has a current collector having one surface and the other surface on which the positive electrode active material layer and the negative electrode active material layer are formed respectively, and has an uncoated portion on which each of the positive electrode active material layer and the negative electrode active material layer is not formed, the uncoated portion of one of the positive-electrode termination electrode and the negative-electrode termination electrode extends in a stacking direction of the stacked electrode assembly together with the separator adjacent to the uncoated portion of the one of the positive-electrode termination electrode and the negative-electrode termination electrode, and is folded so as to overlap with the other of the positive-electrode termination electrode and the negative-electrode termination electrode with the separator being interposed between the uncoated portion of the one of the positive-electrode termination electrode and the negative-electrode termination electrode and the other of the positive-electrode termination electrode and the negative-electrode termination electrode, and the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode extend in the stacking direction of the stacked electrode assembly together with the separator adjacent to each of the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode, and are folded so as to be stacked on the one of the positive-electrode termination electrode and the negative-electrode termination electrode with the separator being interposed between each of the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode and the one of the positive-electrode termination electrode and the negative-electrode termination electrode.

2. The power storage device according to claim 1, wherein a thickness of the separator in a region folded so as to be stacked on the one of the positive-electrode termination electrode and the negative-electrode termination electrode is thinner than a thickness of the separator in a region included in the stacked electrode assembly.

3. A vehicle on which the power storage device according to claim 1 is mounted, wherein the power storage device is mounted on a floor of the vehicle, and the power storage device is disposed such that a side of the power storage device on which the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode are folded to be stacked on the one of the positive-electrode termination electrode and the negative-electrode termination electrode with the separator being interposed between the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode faces a road surface.

4. A vehicle on which the power storage device according to claim 2 is mounted, wherein the power storage device is mounted on a floor of the vehicle, and the power storage device is disposed such that a side of the power storage device on which the uncoated portion of the other of the

positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode are folded to be stacked on the one of the positive-electrode termination electrode and the negative-electrode termination electrode with the separator being interposed between the uncoated portion of the other of the positive-electrode termination electrode and the negative-electrode termination electrode and the uncoated portion of the bipolar electrode faces a road surface.
